TRANSPORTATION MODELS: LEAST-COST METHOD

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Some special items are produced at factories at Patna, Guwahati, Mumbai and Hyderabad. Distributors are located at Lucknow, Delhi, Jaipur, Chennai and Kolkata. Supply and demands of items are indicated in table. Note that factories have capacity to produce 18,000 items, but only 15,000 items are in demand. Transportation cost of each item is proportional to distance it travels. The road distances from factory to distribution centers (in km) are indicated in each cell. Your goal is to minimize the transportation cost.

Distributor Factory	Lucknow	Delhi	Jaipur	Chennai	Kolkata	Supply
Patna	610	1,060	1,180	1,490	580	3,000
Guwahati	1,020	1,910	1,946	2,700	1,018	6,000
Mumbai	1,370	1,420	1,150	1,330	2,050	4,000
Hyderabad	1,300	1,560	1,470	630	1,500	5,000
Demand	2,000	7,000	4,000	1,000	1,000	

Unbalanced Problem: add dummy destination

Add a dummy demand constraint with 0 unit cost and with allocation (18000 – 15000) = 3000.

First allocate cells except dummy column and at last step in dummy column.

Distributor Factory	Lucknow	Delhi	Jaipur	Chennai	Kolkata	Dummy	Supply
Patna	610	1,060	1,180	1,490	580	0	3,000
Guwahati	1,020	1,910	1,946	2,700	1,018	0	6,000
Mumbai	1,370	1,420	1,150	1,330	2,050	0	4,000
Hyderabad	1,300	1,560	1,470	630	1,500	0	5,000
Demand	2,000	7,000	4,000	1,000	1,000	3000	

Stepwise Procedure

Step 1

The least unit cost is 580 in cell (Patna, Kolkata). Check corresponding Supply (3000) and Demand (1000) values. The allocation to this cell is min (3000, 1000) = 1000. This satisfies the entire demand of Kolkata (cross out the column Kolkata) and leaves 3000 – 1000 = 2000 units with Patna.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610	1060	1180	1490	580(1000)	0	2000
Guwahati	1020	1910	1946	2700	1018	0	6000
Mumbai	1370	1420	1150	1330	2050	0	4000
Hyderabad	1300	1560	1470	630	1500	0	5000
Demand	2000	7000	4000	1000	0	3000	

Step 2

The next least unit cost is 610 in cell (Patna, Lucknow). Check corresponding Supply (2000) and Demand (2000) values. The allocation to this cell is min (2000, 2000) = 2000.

This meets the complete demand of Lucknow (cross out the column Lucknow) and leaves 2000 – 2000 = 0 unit with Patna (cross out the row Patna).

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610(2000)~~	~~1060~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0
Guwahati	1020	1910	1946	2700	1018	0	6000
Mumbai	1370	1420	1150	1330	2050	0	4000
Hyderabad	1300	1560	1470	630	1500	0	5000
Demand	0	7000	4000	1000	0	3000	

Step 3

The least unit cost is 630 in cell (Hyderabad, Chennai). Check corresponding Supply (5000) and Demand (1000) values. The allocation to this cell is min (5000, 1000) = 1000.

This satisfies the entire demand of Chennai (cross out the column Chennai) and leaves 5000 – 1000 = 4000 units with Hyderabad.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610(2000)~~	~~1060~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0
Guwahati	1020	1910	1946	2700	1018	0	6000
Mumbai	1370	1420	1150	1330	2050	0	4000
Hyderabad	1300	1560	1470	630(1000)	1500	0	4000
Demand	0	7000	4000	0	0	3000	

Step 4

The least unit cost is 1150 in cell (Mumbai, Jaipur). Check corresponding Supply (4000) and Demand (4000) values. The allocation to this cell is min (4000, 4000) = 4000.

This meets the complete demand of Jaipur (cross out the column Jaipur) and leaves 4000 – 4000 = 0 unit with Mumbai (cross out the row Mumbai).

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610(2000)~~	~~1060~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0
Guwahati	1020	1910	1946	2700	1018	0	6000
~~Mumbai~~	~~1370~~	~~1420~~	~~1150(4000)~~	~~1330~~	~~2050~~	~~0~~	0
Hyderabad	1300	1560	1470	630(1000)	1500	0	4000
Demand	0	7000	0	0	0	3000	

Step 5

The least unit cost is 1560 in cell (Hyderabad, Delhi). Check corresponding Supply (4000) and Demand (7000) values. The allocation to this cell is min (4000, 7000) = 4000.

This leaves 7000 – 4000 = 3000 units with Delhi and 4000 – 4000 = 0 unit with Hyderabad (cross out the row Hyderabad).

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610(2000)~~	~~1060~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0
Guwahati	1020	1910	1946	2700	1018	0	6000
~~Mumbai~~	~~1370~~	~~1420~~	~~1150(4000)~~	~~1330~~	~~2050~~	~~0~~	0
~~Hyderabad~~	~~1300~~	~~1560(4000)~~	~~1470~~	~~630(1000)~~	~~1500~~	~~0~~	0
Demand	0	3000	0	0	0	3000	

Step 6

The least unit cost is 1910 in cell (Guwahati, Delhi). Check corresponding Supply (6000) and Demand (3000) values. The allocation to this cell is min (6000, 3000) = 3000. This satisfies the entire demand of Delhi (cross out the column Delhi) and leaves 6000 – 3000 = 3000 units with Guwahati.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610(2000)~~	~~1060~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0
Guwahati	1020	1910(3000)	1946	2700	1018	0	3000
~~Mumbai~~	~~1370~~	~~1420~~	~~1150(4000)~~	~~1330~~	~~2050~~	~~0~~	0
~~Hyderabad~~	~~1300~~	~~1560(4000)~~	~~1470~~	~~630(1000)~~	~~1500~~	~~0~~	0
Demand	0	0	0	0	0	3000	

Step 7

The least unit cost is 0 in cell (Guwahati, D_{dummy}). Check corresponding Supply (3000) and Demand (3000) values. The allocation to this cell is min (3000, 3000) = 3000. This satisfies the entire demand of D_{dummy} (cross out the column D_{dummy}) and leaves 3000 – 3000 = 0 unit with Guwahati (cross out the column Guwahati).

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610(2000)~~	~~1060~~	~~1180~~	~~1490~~	~~580(1000)~~	~~0~~	0
~~Guwahati~~	~~1020~~	~~1910(3000)~~	~~1946~~	~~2700~~	~~1018~~	~~0(3000)~~	0
~~Mumbai~~	~~1370~~	~~1420~~	~~1150(4000)~~	~~1330~~	~~2050~~	~~0~~	0
~~Hyderabad~~	~~1300~~	~~1560(4000)~~	~~1470~~	~~630(1000)~~	~~1500~~	~~0~~	0
Demand	0	0	0	0	0	0	

Step 8

We have reached the final initial allocation. The initial feasible solution is

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610 **(2000)**	1060	1180	1490	580 **(1000)**	0	3000
Guwahati	1020	1910 **(3000)**	1946	2700	1018	0 **(3000)**	6000
Mumbai	1370	1420	1150 **(4000)**	1330	2050	0	4000
Hyderabad	1300	1560 **(4000)**	1470	630 **(1000)**	1500	0	5000
Demand	2000	7000	4000	1000	1000	3000	

The minimum total transportation cost

$= 610 \times 2000 + 580 \times 1000 + 1910 \times 3000 + 0 \times 3000 + 1150 \times 4000 + 1560 \times 4000 + 630 \times 1000$

$= 19{,}000{,}000.$

- **Least-cost method:**

This method finds a better starting solution by targeting the cheapest routes.

It assigns as much as possible to the cell with the smallest unit cost (ties are broken arbitrarily)

Next, the satisfied row or column is crossed out and the amounts of supply and demand are adjusted accordingly.

If both row and a column are satisfied simultaneously, only one is crossed out, the same as in the northwest-corner method. Next, select the uncrossed-out cell with the smallest unit cost and repeat the process until exactly one row or column is left uncrossed out.

(Based on book by H.A. Taha).

- **Benefits of Least-cost method:**

- This method provides better initial solution than Northwest Corner Method
- Simple and intuitive
- May reduces iteration in optimization (Modified Distribution Method)

- **Limitations of Least-cost method:**

- Does not guarantee optimality

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610 **(2000)**	1060	1180	1490	580 **(1000)**	0	3000
Guwahati	1020 (0)	1910 **(3000)**	1946	2700	1018 (0)	0 **(3000)**	6000
Mumbai	1370	1420	1150 **(4000)**	1330	2050	0	4000
Hyderabad	1300	1560 **(4000)**	1470	630 **(1000)**	1500	0	5000
Demand	2000	7000	4000	1000	1000	3000	

	D1	D2	D3	D4	D5	Supply
S1	3	4	6	8	9	30
S2	2	2	4	5	5	80
S3	2	2	2	3	2	10
S4	3	3	2	4	2	20
Demand	10	50	20	40	20	

	D_1	D_2	D_3	D_4	D_5	Supply
S_1	3	4	6	8	9	30
S_2	2(10)	2(50)	4	5	5	20
~~S_3~~	~~2~~	~~2~~	2(10)	~~3~~	~~2~~	0
~~S_4~~	~~3~~	~~3~~	~~2~~	~~4~~	2(20)	0
Demand	0	0	10	40	0	

	D_1	D_2	D_3	D_4	D_5	Supply
S_1	3	4	6	8	9	30
S_2	2(10)	2(50)	4(10)	5	5	10
S_3	2	2	2(10)	3	2	0
S_4	3	3	2	4	2(20)	0
Demand	0	0	0	40	0	

	D_1	D_2	D_3	D_4	D_5	Supply
S_1	3	4	6	8	9	30
~~S_2~~	~~2~~**~~(10)~~**	~~2~~**~~(50)~~**	~~4~~**~~(10)~~**	~~5~~**~~(10)~~**	~~5~~	0
~~S_3~~	~~2~~	~~2~~	~~2~~**~~(10)~~**	~~3~~	~~2~~	0
~~S_4~~	~~3~~	~~3~~	~~2~~	~~4~~	~~2~~**~~(20)~~**	0
Demand	0	0	0	30	0	

	D_1	D_2	D_3	D_4	D_5	Supply
S_1	3	4	6	8(30)	9	0
S_2	2(10)	2(50)	4(10)	5(10)	5	0
S_3	2	2	2(10)	3	2	0
S_4	3	3	2	4	2(20)	0
Demand	0	0	0	0	0	

Minimum total transportation cost =
$8\times30+2\times10+2\times50+4\times10+5\times10+2\times10+2\times20 = 510$